

Problem

A parallel RLC circuit has the node equation

$$\frac{dv}{dt} + 50v + 100 \int v dt = 110 \cos(377t - 10^\circ) \text{ V}$$

Determine $v(t)$ using the phasor method. You may assume that the value of the integral at $t = -\infty$ is zero.

Step-by-step solution

Step 1 of 1

$$\frac{dV}{dt} + 50V + 100 \int V dt = 110 \cos(377t - 10^\circ)$$

Assuming integral value at $t = -\infty$ is given as zero.

Representing in frequency domain,

$$j\omega V + 50V + \frac{100V}{j\omega} = 110 \angle -10^\circ$$

$$\therefore \omega = 377$$

$$V(j377 + 50 - j0.265) = 110 \angle -10^\circ$$

$$V(50 + j376.735) = 110 \angle -10^\circ$$

$$V = \frac{110 \angle -10^\circ}{380.038 \angle 82.439^\circ}$$

$$V = 0.289 \angle -92.439^\circ$$

$$\text{In time domain, } V(t) = 0.289 \cos(377t - 92.439^\circ)$$